Software Requirements Specification

Version 1.0
<<Annotated Version>>

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Smart Inventory and Sales Analytics System
for Small Businesses

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Submitted in partial fulfillment
Of the requirements of
DSC-11 Software Engineering

<<Any comments inside double brackets such as these are *not* part of this SRS but are comments upon this SRS example to help the reader understand the point being made.

Refer to the SRS Template for details on the purpose and rules for each section of this document.

This work is based upon the submissions of the Spring 2004 CS 310. The students who submitted these team projects were Atharv Sinha, Arpit Verma, Piyush Jaiswal, Dheeraj, Rahul Kumar. >>

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1.0. Introduction

1.1. Purpose

The purpose of this document is to present a detailed description of the Smart Inventory and Sales Analytics System for Small Businesses. It will explain the purpose and features of the system, the interfaces of the system, what the system will do, the constraints under which it must operate and how the system will react to external stimuli. This document is intended for both the stakeholders and the developers of the system and will be proposed to Dr. Rekha for its approval.

1.2. Scope of Project

The scope of the Smart Inventory and Sales Analytics System for Small Businesses is to provide a simple, efficient, and reliable software solution for managing inventory, sales, and basic business analytics for small-scale enterprises such as kirana stores, retail shops, pharmacies, and wholesalers.

The system will allow users to manually enter product, stock, and sales data through an easy-to-use interface designed for users with minimal technical knowledge. It will maintain real-time stock records, track product-wise sales, and automatically calculate daily, monthly, and yearly sales summaries.

The project includes an integrated billing module that enables fast and accurate invoice generation during sales transactions. All sales data will be stored and used for generating analytical reports, including sales trends and product performance insights.

1.3. Glossary

Term	Definition
Inventory	The complete list of products or goods stored by a business for sale.
Stock Level	The quantity of a particular product currently available in the store.
Billing System	A module used to generate invoices and record sales transactions..
Invoice	A receipt generated after completion of a sales transaction.
Sales Analytics	The process of analyzing sales data to identify trends and product performance.
Sales Trend	Pattern showing increase or decrease in product sales over time.
Stale Inventory	Items that have not been sold for a long period of time.
Expiring Stock	Products approaching their expiry date, especially relevant in pharmacies.
Medicine Composition	Active ingredients or salts present in a medicine.
Alternative Medicine	Substitute medicine with similar composition and usage.
Software Requirements Specification	A document that completely describes all of the functions of a proposed system and the constraints under which it must operate. For example, this document.
Stakeholder	Any person with an interest in the project who is not a developer.
User	Shopkeeper or Pharmacist

1.4. References

IEEE. *IEEE Std 830-1998 IEEE Recommended Practice for Software Requirements Specifications.* IEEE Computer Society, 1998.

1.5. Overview of Document

The next chapter, the Overall Description section, of this document gives an overview of the functionality of the product. It describes the informal requirements and is used to establish a context for the technical requirements specification in the next chapter.

The third chapter, Requirements Specification section, of this document is written primarily for the developers and describes in technical terms the details of the functionality of the product.

Both sections of the document describe the same software product in its entirety, but are intended for different audiences and thus use different language.

2.0. Overall Description

2.1 System Environment

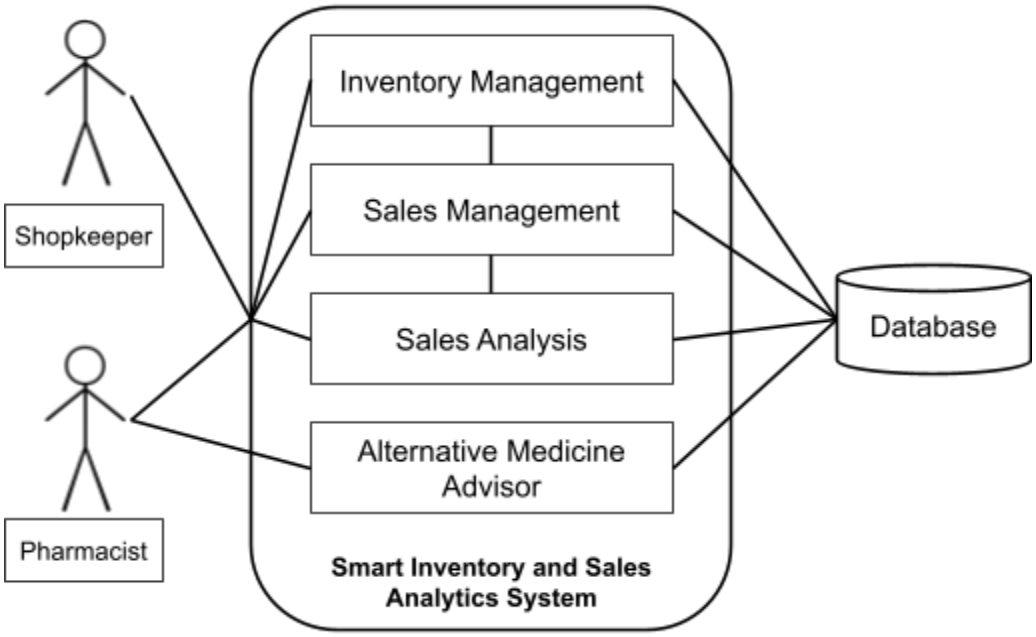


Figure 1 - System Environment

The Smart Inventory and Sales Analytics System consists of two active actors: the Retail Shopkeeper and the Pharmacist. The Retail Shopkeeper uses the system to manage inventory, perform billing operations, record sales transactions, and view sales analysis reports for general retail products. The Pharmacist uses the system in a manner similar to the Retail shopkeeper but also accesses the Alternative Medicine Advisor to identify suitable substitute medicines in case of non-availability.

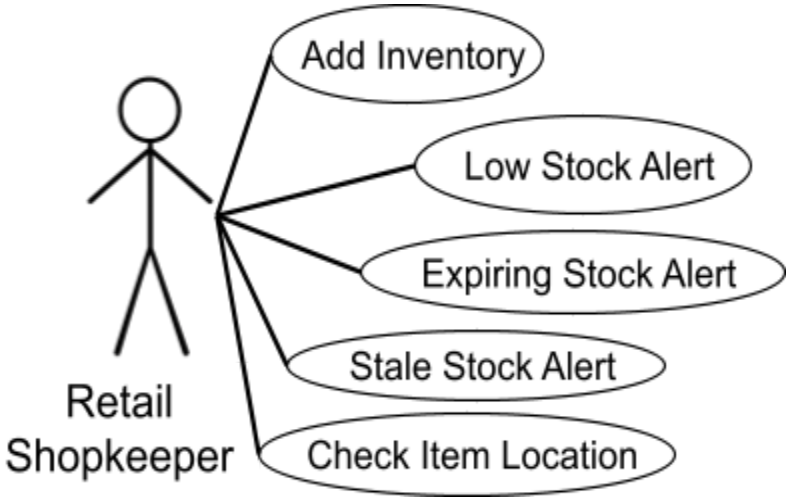
The system includes core modules such as Inventory Management, Sales Management, Sales Analysis, and the Alternative Medicine Advisor. All user interactions are processed within the system boundary, and relevant data is stored and retrieved from a cloud database.

2.2 Functional Requirements Specification

This section outlines the use cases for each of the active users separately. The Retail Shopkeeper has 3 use cases while the Pharmacist has 4 use cases.

2.2.1 Retail Shopkeeper Use Case

Use case: **Inventory Management**
Diagram:



Brief Description
The Retail Shopkeeper accesses the the Smart Inventory and Sales Analytics System, uploads info about new stock(name, quantity, purchase price per piece, storage location, expiry date); looks up the location, quantity or exp date of a specific product; gets automatic alerts for low, expiring or stale stock(in notifications dropbox).

Initial Step-By-Step Description
Before this use case can be initiated, the Retail Shopkeeper has already accessed The Smart Inventory and Sales Analytics System.

➢ **To Add new inventory:**
1. The Retail Shopkeeper accesses the add/update inventory menu from the Inventory management tab.
2. The system displays forms for adding/updating inventory.
3. The Retail Shopkeeper enters corresponding data and presses submit.
4. The system saves the entered data into the database.
5. The system displays a success message or error message depending on the output.

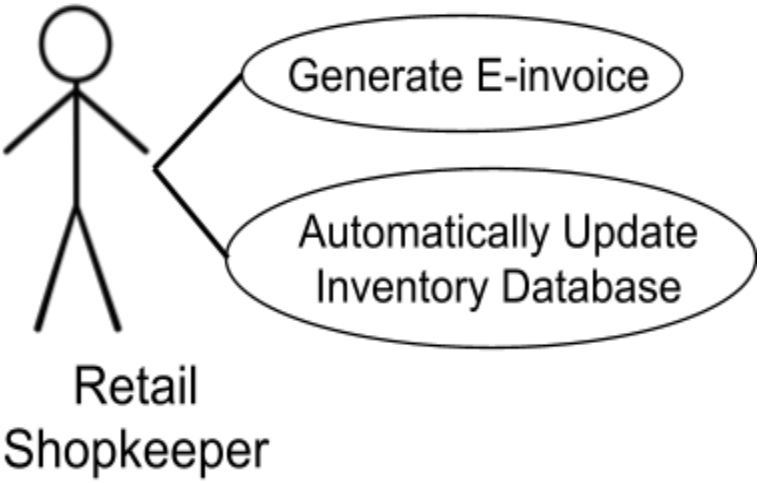
➢ **To Check Item Status:**
1. The Retail Shopkeeper accesses the check item menu from the Inventory management tab.

2. The system asks for the item name.
3. The Retail Shopkeeper enters the item name.
4. The system displays the data from the database.

➢ **To check Alerts:**
1. The system automatically saves and displays alerts in the notifications menu.
2. The Retail Shopkeeper checks alerts from the notifications menu.

Xref: Section 3.2.1, Inventory Management

Use case: **Sales Management**
Diagram:



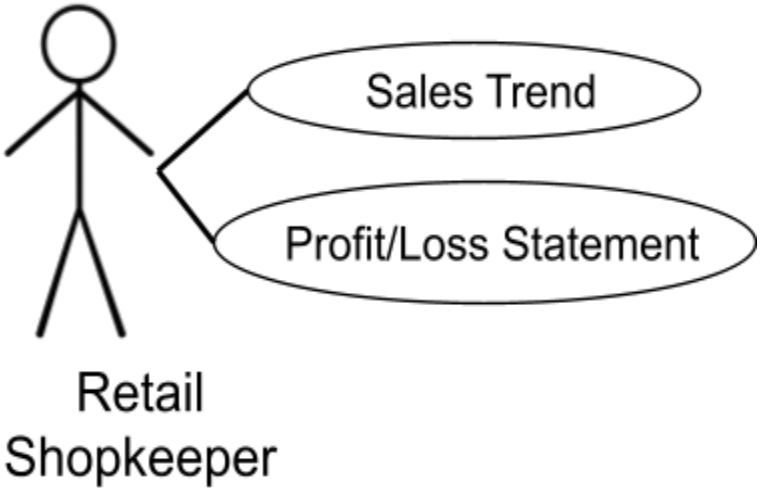
Brief Description
The Retail Shopkeeper accesses the Smart Inventory and Sales Analytics System, enters the name of products being bought by the customer and its sales price(in case of discount) to generate E-invoice.

Initial Step-By-Step Description
Before this use case can be initiated, the Retail Shopkeeper has already accessed The Smart Inventory and Sales Analytics System.

1. The Retail Shopkeeper accesses the Sales management tab.
2. The system displays forms for entering the product name and sales price.
3. The Retail Shopkeeper enters the product name.
4. The system displays the usual sales price and mrp for that product.
5. The Retail Shopkeeper enters the sales price.
6. The Retail Shopkeeper repeats steps 3-5 for other products.
7. The Retail Shopkeeper presses the generate bill button.
8. The system generates the E-invoice.

Xref: Section 3.2.2, Sales Management

Use case: **Sales Analysis**
Diagram:



Brief Description
The Retail Shopkeeper accesses the the Smart Inventory and Sales Analytics System; views the month's/year's product sales trends with correlation to changes in months/seasons; views the Profit/Loss Statement, which includes the quantities of products sold, the amount used to purchase them and the amount sold at.

Initial Step-By-Step Description
Before this use case can be initiated, the Retail Shopkeeper has already accessed The Smart Inventory and Sales Analytics System.

➢ **To View Sales Trend:**
1. The Retail Shopkeeper accesses the Sales Trends menu from the Sales Analysis tab.
2. The Retail Shopkeeper chooses the type of trend they want to see(month/year; overall/specific product).
3. The system displays the graphical trend as requested.
4. The system correlates the graphical data with the changes in months/seasons or specific festivals.

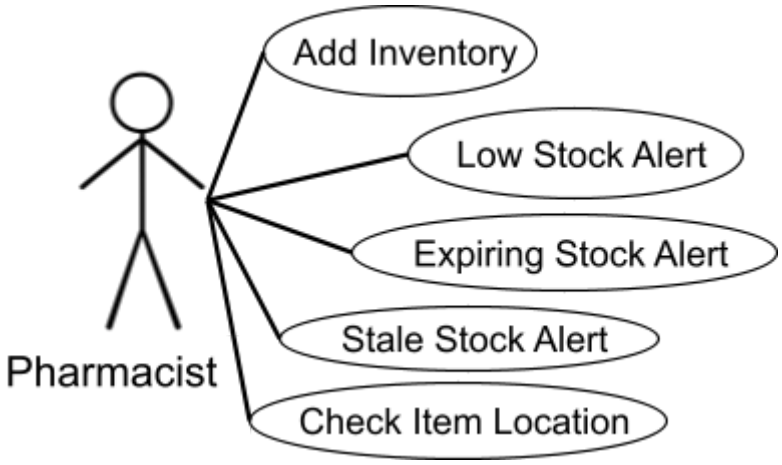
➢ **To View Profit/Loss Statement:**
1. The Retail Shopkeeper accesses the Profit/Loss Statement menu from the Sales Analysis tab.
2. The Retail Shopkeeper chooses the duration of the statement they want to see(month/year).
3. The system displays the month's/year's profit/loss statement.
4. The statement displays the profit/loss based on the amount spent to purchase the products sold or discarded(due to expiration) and the amount they were sold at.

5. The statement contains a dropbox with detailed records of quantities of each product sold or discarded, the overall amount used to purchase them and the amount they were sold at.

Xref: Section 3.2.3, Sales Analysis

2.2.2 Pharmacist Use Case

Use case: **Inventory Management**
Diagram:



Brief Description
The Pharmacist accesses the the Smart Inventory and Sales Analytics System, uploads info about new stock(name, quantity, purchase price per piece, storage location, expiry date, composition); looks up the location, quantity or exp date of a specific product; gets automatic alerts for low, expiring or stale stock(in notifications dropbox).

Initial Step-By-Step Description
Before this use case can be initiated, the Pharmacist has already accessed The Smart Inventory and Sales Analytics System.

➢ **To Add new inventory:**
1. The Pharmacist accesses the add/update inventory menu from the Inventory management tab.
2. The system displays forms for adding/updating inventory.
3. The Pharmacist enters corresponding data and presses submit.
4. The system saves the entered data into the database.
5. The system displays a success message or error message depending on the output.

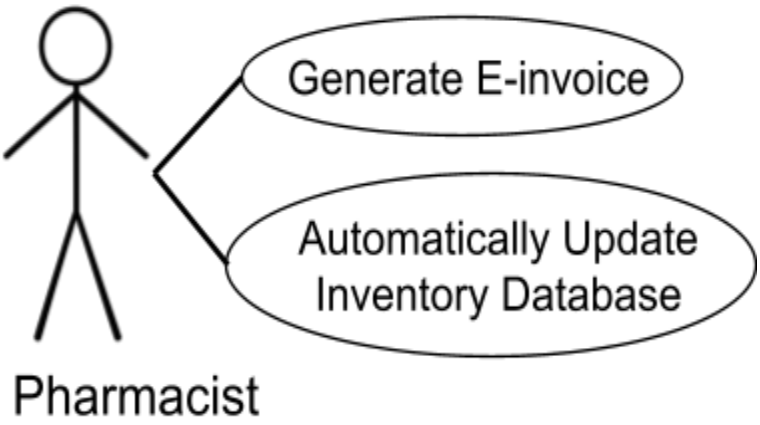
➢ **To Check Item Status:**
1. The Pharmacist accesses the check item menu from the Inventory management tab.
2. The system asks for the item name.

3. The Pharmacist enters the item name.
4. The system displays the data from the database.

➢ **To check Alerts:**
1. The system automatically saves and displays alerts in the notifications menu.
2. The Pharmacist checks alerts from the notifications menu.

Xref: Section 3.2.1, Inventory Management

Use case: **Sales Management**
Diagram:



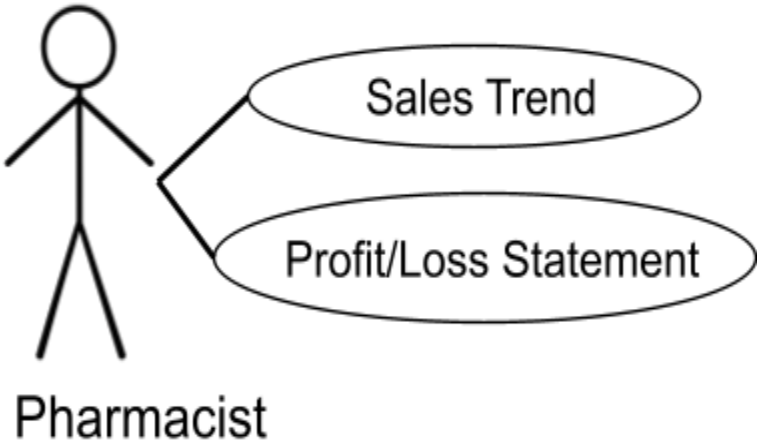
Brief Description
The Pharmacist accesses the Smart Inventory and Sales Analytics System, enters the name of products being bought by the customer and its sales price(in case of discount) to generate E-invoice.

Initial Step-By-Step Description
Before this use case can be initiated, the Pharmacist has already accessed The Smart Inventory and Sales Analytics System.

1. The Pharmacist accesses the Sales management tab.
2. The system displays forms for entering the product name and sales price.
3. The Pharmacist enters the product name.
4. The system displays the usual sales price and mrp for that product.
5. The Pharmacist enters the sales price.
6. The Pharmacist repeats steps 3-5 for other products.
7. The Pharmacist presses the generate bill button.
8. The system generates the E-invoice.

Xref: Section 3.2.2, Sales Management

Use case: **Sales Analysis**
Diagram:



Brief Description
The Pharmacist accesses the the Smart Inventory and Sales Analytics System; views the month's/year's product sales trends with correlation to changes in months/seasons; views the Profit/Loss Statement, which includes the quantities of products sold, the amount used to purchase them and the amount sold at.

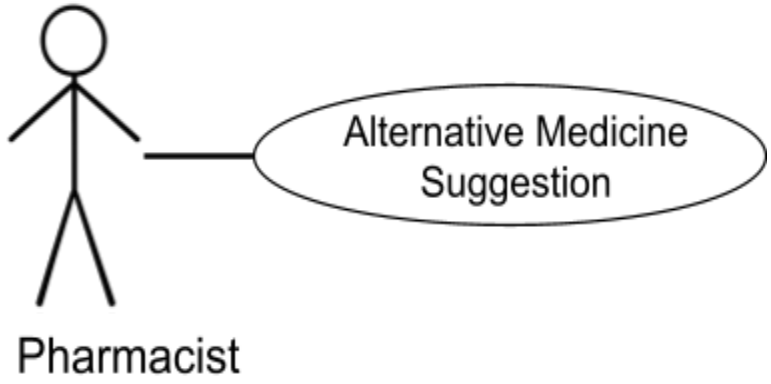
Initial Step-By-Step Description
Before this use case can be initiated, the Pharmacist has already accessed The Smart Inventory and Sales Analytics System.

➢ **To View Sales Trend:**
1. The Pharmacist accesses the Sales Trends menu from the Sales Analysis tab.
2. The Pharmacist chooses the type of trend they want to see(month/year; overall/specific product).
3. The system displays the graphical trend as requested.
4. The system correlates the graphical data with the changes in months/seasons or specific festivals.

➢ **To View Profit/Loss Statement:**
1. The Pharmacist accesses the Profit/Loss Statement menu from the Sales Analysis tab.
2. The Pharmacist chooses the duration of the statement they want to see(month/year).
3. The system displays the month's/year's profit/loss statement.
4. The statement displays the profit/loss based on the amount spent to purchase the products sold or discarded(due to expiration) and the amount they were sold at.
5. The statement contains a dropbox with detailed records of quantities of each product sold or discarded, the overall amount used to purchase them and the amount they were sold at.

Xref: Section 3.2.3, Sales Analysis

Use case: **Alternative Medicine Advisor**
Diagram:



Brief Description
The Pharmacist accesses the Smart Inventory and Sales Analytics System, enters the composition of medicine needed by the customer and gets the names of alternative medicines containing that composition.

Initial Step-By-Step Description
Before this use case can be initiated, the Pharmacist has already accessed The Smart Inventory and Sales Analytics System.

1. The Pharmacist accesses the Alternative Medicine tab.
2. The system displays forms for entering the medicine composition.
3. The Pharmacist enters the composition and presses search.
4. The system displays alternative medicines and their full compositions.

Xref: Section 3.2.4, Alternative Medicine Advisor

<< Since both of the actors only have similar use cases and required software modules, which is the same as the system environment, the summary diagram has been omitted as it will be the same as the system environment diagram.>>

2.3 User Characteristics

The Retail Shopkeeper is expected to have basic computer literacy and be able to operate a desktop-based application. The user should be capable of entering product

details, recording sales transactions, and viewing inventory and sales reports through the system interface.

The Medical Store Shopkeeper is expected to have basic computer knowledge and familiarity with medicine names and compositions. The user should be able to manage medicine inventory, check expiry information, and use the alternative medicine suggestion feature when required.

Both users are expected to understand basic shop operations and should be able to use standard interface elements such as buttons, forms, and menus. The system is designed to be simple and requires minimal training for effective usage.

2.4 Non-Functional Requirements

The Smart Inventory and Sales Analytics System will operate as a cloud-based application. The system will use a cloud-hosted database so that users are not required to install or configure any local database management system.

The software will be accessible through an internet-enabled device and will run on standard operating systems using a supported web browser. System performance will depend on the user's internet connectivity and device capabilities.

All inventory and sales data will be stored securely in the cloud database to ensure data availability, reliability, and centralized data management.

3.0. Requirements Specification

3.1 External Interface Requirements

The Smart Inventory and Sales Analytics System interfaces with a cloud-hosted database for storing inventory, sales, and user-related data. This external database provides centralized data storage and eliminates the need for local database setup by the user.

The system communicates with the database through secure backend services to store and retrieve product, and sales information. Users access the system through a standard web browser, which serves as the primary interface between the user and the application.

No other external systems are required for normal system operation.

3.2 Functional Requirements

The Logical Structure of the Data is contained in Section 3.3.1.

3.2.1 Inventory Management

Use Case Name	Inventory Management
XRef	Section 2.2.1, Inventory Management Section 2.2.2, Inventory Management
Trigger	The User accesses the Inventory Management tab.
Precondition	The Web is displayed with the home page containing the different tabs.
Basic Path	1. The user chooses which feature to use. The choices are: ‘add inventory’ and ‘check item status’. 2. If the user chooses ‘add inventory’, the system displays forms for adding/updating inventory. On submitting the form, the system saves the entered data into the database. 3. If the user chooses ‘check item status’, the system displays a form to enter the item name. On submitting the form, the system item information(name, quantity, purchase price per piece, storage location, expiry date, composition, etc.) from the database.
Postcondition	If in step 2, the database does not store the data and returns an error, the system asks the user to fix the error and resubmit.

Exception Paths	The user may abandon the process at any time
Other	The item status is generated from the information provided when the user adds to or updates the inventory and not predefined in the database.

3.2.2 Sales Management

Use Case Name	Sales Management
XRef	Section 2.2.1, Sales Management Section 2.2.2, Sales Management
Trigger	The User accesses the Sales Management tab.
Precondition	The Web is displayed with the home page containing the different tabs
Basic Path	1. The system displays forms for entering the item name and sales price. 2. The user enters the product name. The system displays the usual sales price and mrp for that item. The user enters the sales price. The user repeats this for the other products. 3. The user presses the generate bill button. The system generates the E-invoice.
Postcondition	The user may share the invoice with the customer via sms, email or by printing a hardcopy.
Exception Paths	The process may be abandoned at any time.
Other	The Item names(for autofill) and sales prices are generated from the information provided when the user adds to or updates the inventory and not predefined in the database.

3.2.3 Sales Analysis

Use Case Name	Sales Analysis
XRef	Section 2.2.1, Sales Analysis Section 2.2.2, Sales Analysis
Trigger	The User accesses the Sales Analysis tab.
Precondition	The Web is displayed with the home page containing the different tabs.
Basic Path	1. The user chooses which report to view. The choices are: ‘sales trend’ and ‘profit/loss statement’. 2. If the user chooses ‘sales trend’, the system allows the user to choose the type of trend they want to see(month/year; overall/specific product). Then the system displays the graphical trend and correlates the graphical data with the changes in months/seasons or specific festivals, as requested. 3. If the user chooses ‘profit/loss statement’, the system allows the user to choose the duration of the statement they want to see(month/year).The system displays the month’s/year’s profit/loss statement.

Postcondition	The user may download the report for sharing with their CA or business advisor.
Exception Paths	The user may abandon the operation at any time.
Other	The Items and corresponding sale quantities and prices are generated from the information provided when the user generates an E-invoice and not predefined in the database.

3.2.4 Alternative Medicine Advisor

Use Case Name	Alternative Medicine Advisor
XRef	Section 2.2.2, Alternative Medicine Advisor
Trigger	The Pharmacist accesses the Alternative Medicine Advisor tab.
Precondition	The Web is accessed with a Pharmacist account and displays the home page containing the different tabs.
Basic Path	1. The system displays forms for entering the medicine composition. 2. The Pharmacist enters the composition and presses search. 3. The system displays alternative medicines(in stock) and their full compositions.
Postcondition	The Pharmacist can suggest and sell an alternative medicine to the customer.
Exception Paths	The Pharmacist may abandon the operation at any time.
Other	The medicines and corresponding compositions are generated from the information provided when the user adds to the inventory and not predefined in the database.

3.3 Detailed Non-Functional Requirements

3.3.1 Logical Structure of the Data

The logical structure of the data to be stored in the internal database is given below.

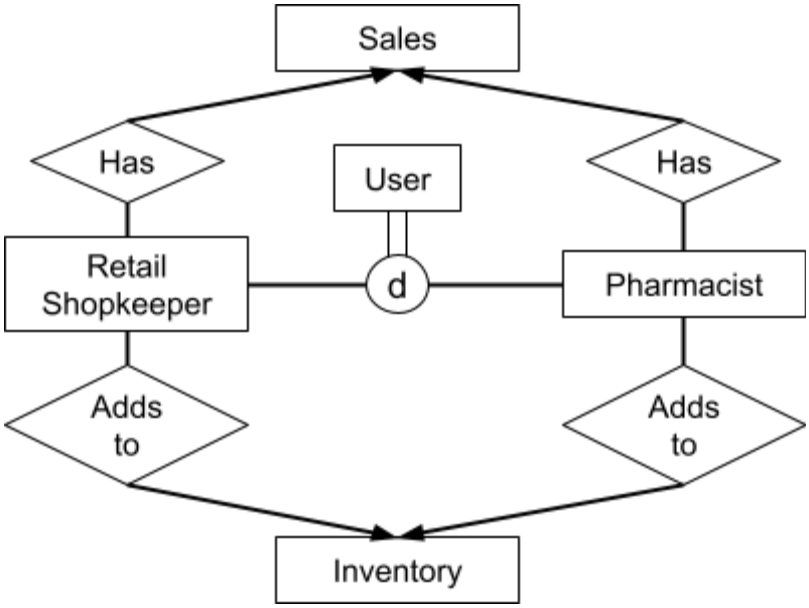


Figure 2 - Logical Structure of the Inventory & Sales Data

The data descriptions of each of these data entities is as follows:

User Entity

Data Item	Type	Description	Comment
User Id	Integer	User Id of Shopkeeper/ Pharmacist	Primary Key
Name	Text	Name of Shopkeeper/ Pharmacist	
IsPharmacist	Boolean	1 for Pharmacist, 0 otherwise	
Email Address	Text	Internet address	
Password	Text	Password	

Inventory Entity(Separate for each user)

Data Item	Type	Description	Comment
Item Name	Text	Name of item	
Batch Id	Text	Batch id of items, may be user defined or from packing details.	Primary Key
Quantity	Integer	Quantity of item purchased	
Purchase price	integer	Purchase price of item(per piece)	
Storage Location	Text	Label of storage location of item	Optional
Expiry Date	Date	Expiry date of item	

Purchase date	Date	Date of batch delivery	Automatically taken from system
Sale Price	Integer	Standard sale price per piece	
MRP	Integer	MRP of item	Optional
Category	Text	Category of item	Optional
Composition	Text	Composition of Medicin	Weak attribute, dependent on user being a Pharmacist (required in this case); Multivalued

Sales Entity(Separate for each user)

Data Item	**Type**	**Description**	**Comment**
Batch Id	Text	Batch id of items	Composite primary key
Month	Integer	Month of item sales	Composite primary key
Year	Integer	Year of item sales	Composite primary key
Sold	Boolean	Sold/discarded; 1 for sold, 0 for discarded	Composite primary key;
Quantity	Integer	Quantity sold/discarded	
Expenditure	Integer	Total amount spent to purchase the sold/discarded items	
Income	Integer	Total amount received on sale	

3.3.2 Security

The Smart Inventory and Sales Analytics System will implement secure user authentication and authorization mechanisms to prevent unauthorized access. Users will be required to log in before accessing system features.

The system will use JSON Web Tokens (JWT) for secure user verification and access control. JWT-based authentication will ensure that only authorized users can perform protected operations such as inventory modification, billing, and data updates.

All application data stored in the cloud database will be protected through server-side access control. Security of the cloud infrastructure and database environment is managed by the hosting service and is considered external to the system.

4.0. Software Design

4.1 System Architecture

The Smart Inventory and Sales Analytics System employs a three-tier client-server architecture. The presentation tier utilizes HTML, CSS, and JavaScript to deliver a responsive web browser interface.

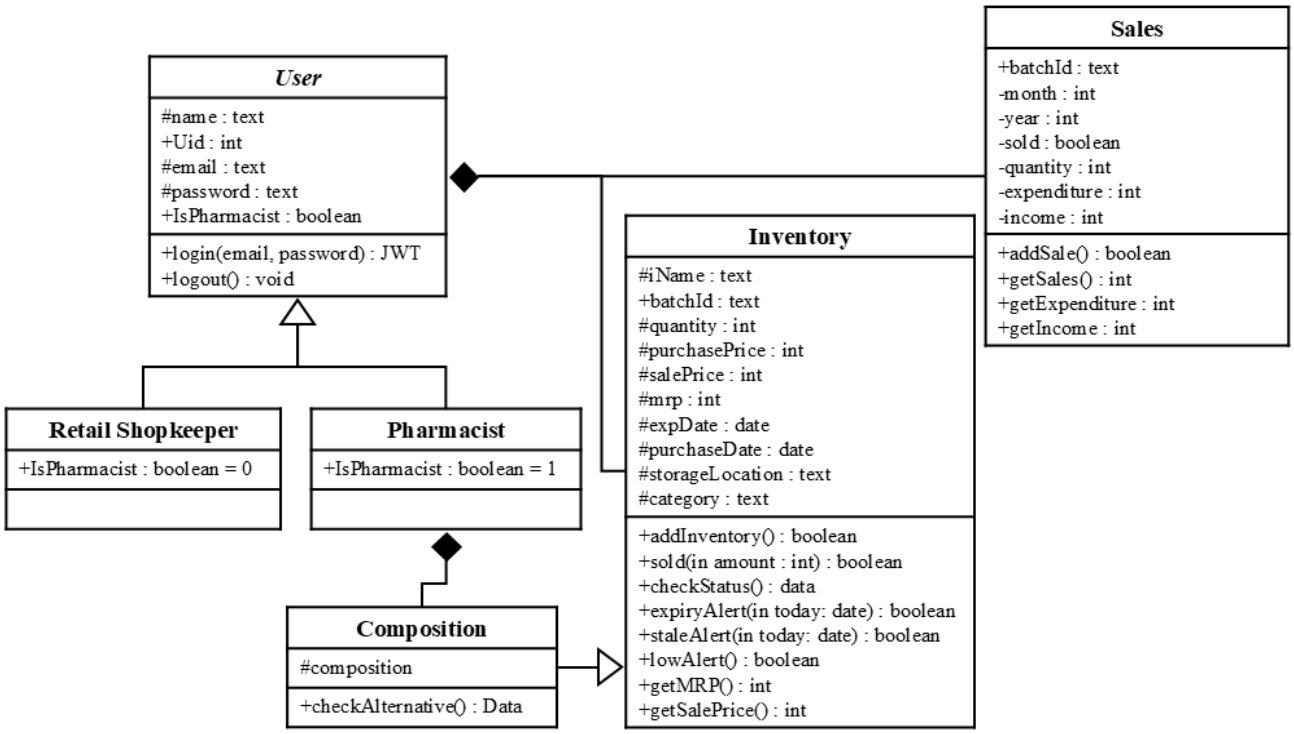
The application tier is driven by a Python backend using the Flask framework, which processes core business logic and system requests. For data persistence, the system connects to a relational MySQL database utilizing the mysql-connector library.

Security protocols are natively integrated into the application tier, employing JSON Web Tokens (JWT) for stateless user authentication and the Bcrypt algorithm for secure password encryption.

4.2 Detailed Design

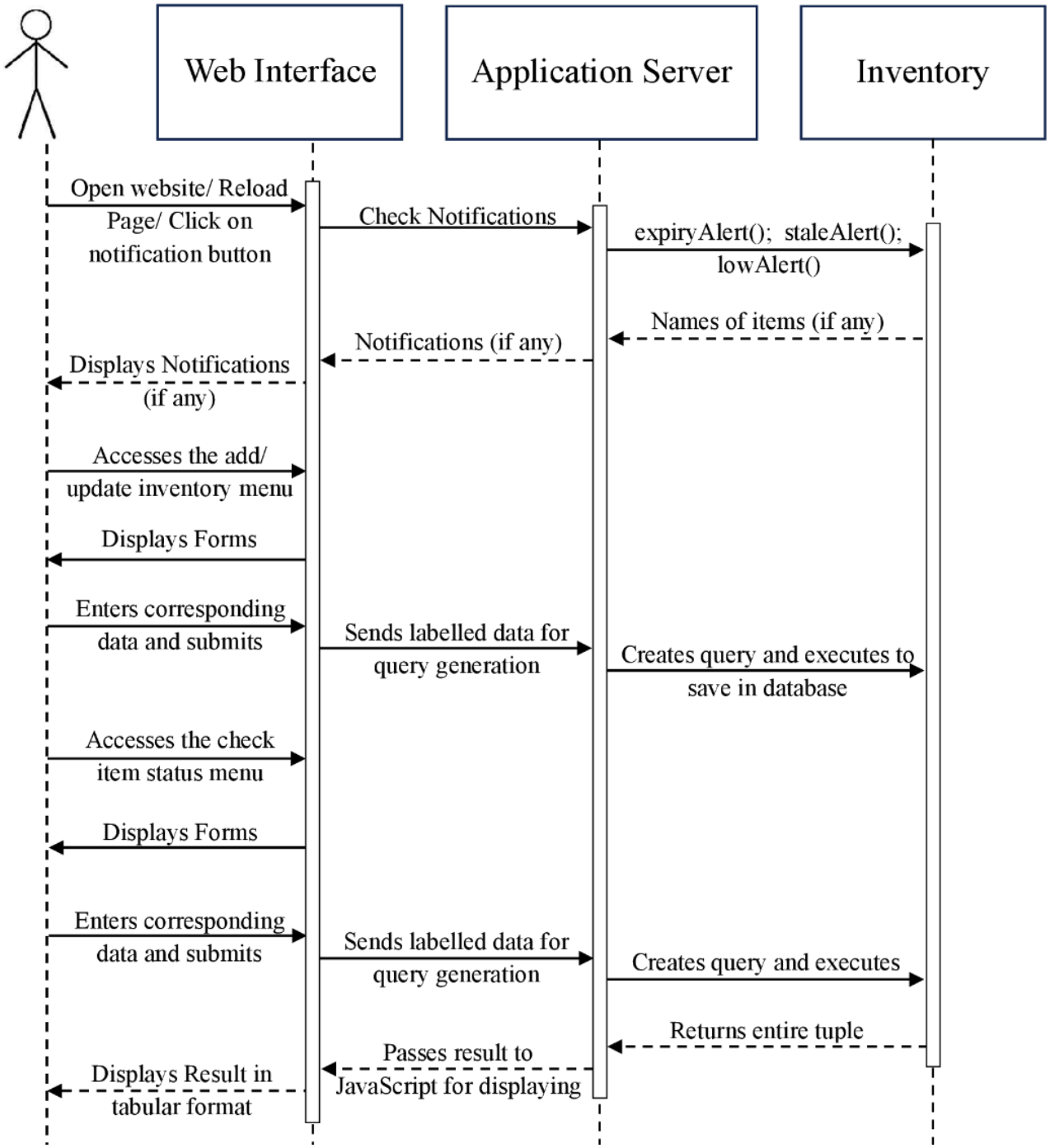
This section outlines the internal structure & interactions of the system components.

4.2.1 Class Diagrams

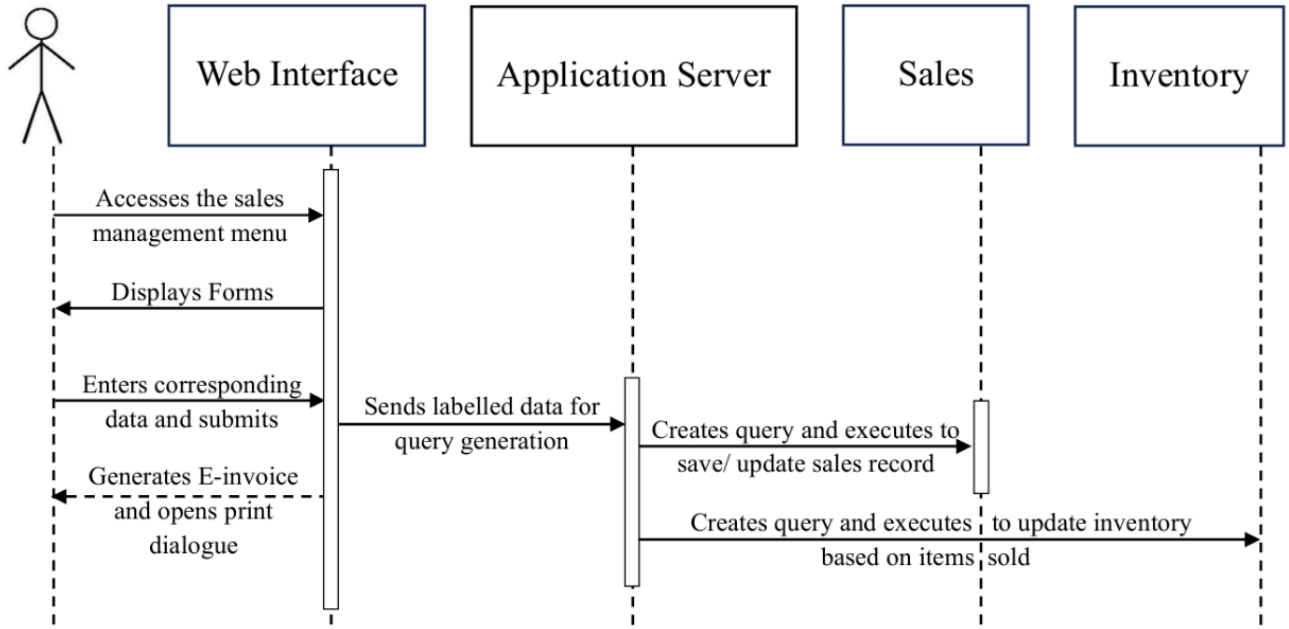


4.2.2 Sequence Diagrams

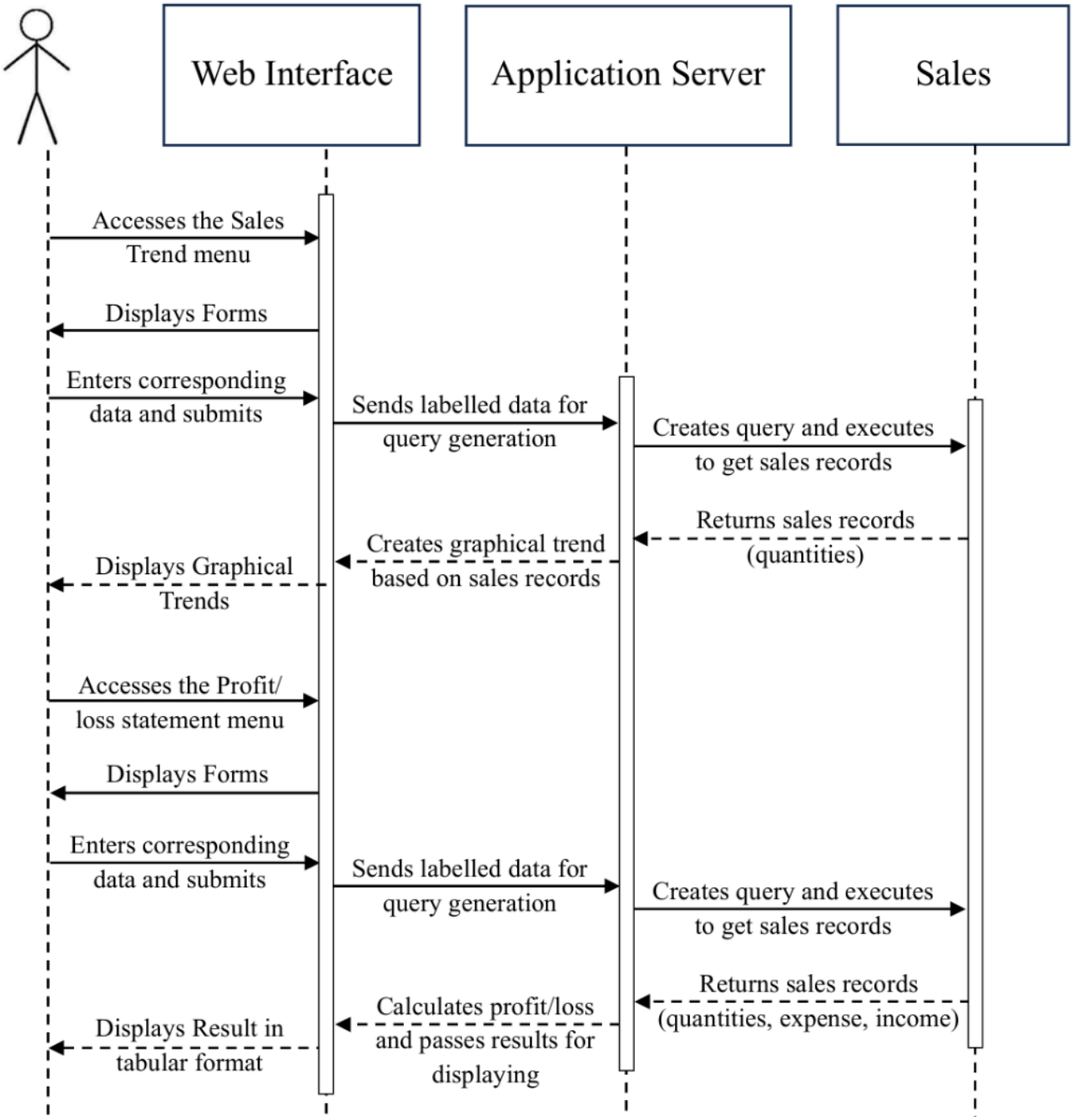
4.2.2.1 Inventory Management



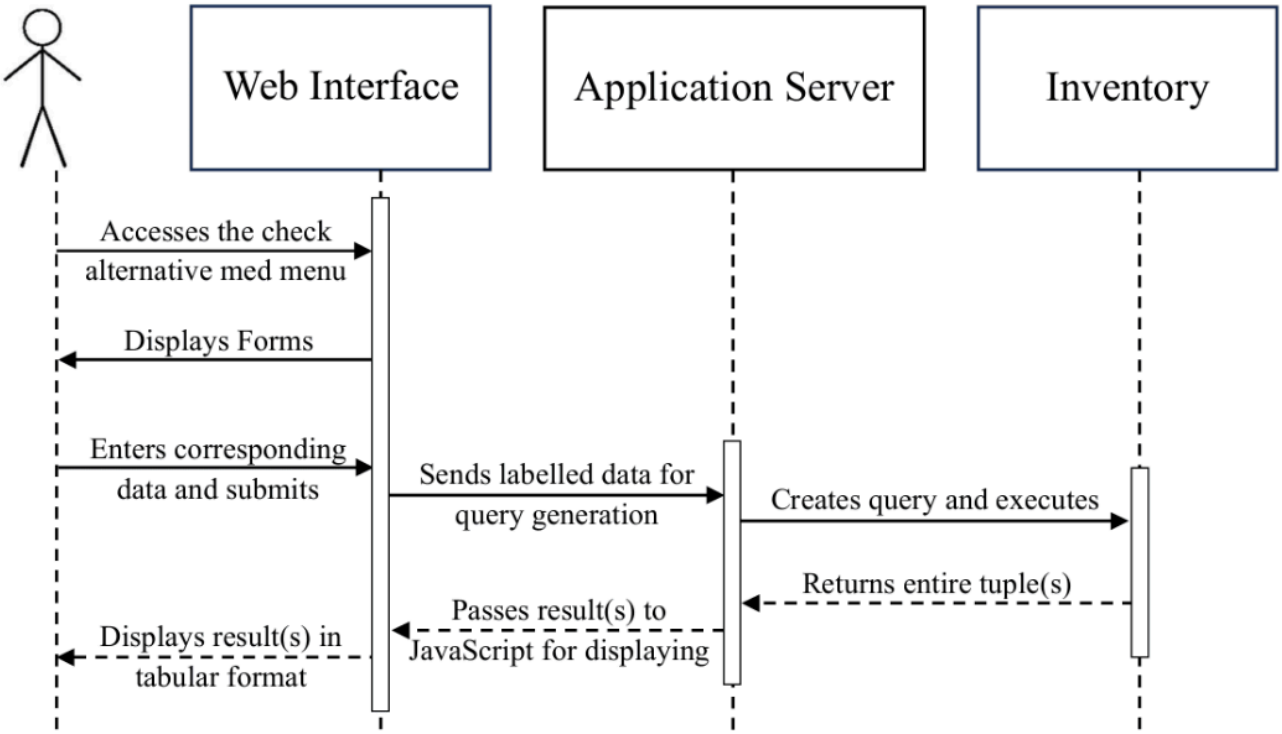
4.2.2.2 Sales Management



4.2.2.3 Sales Analytics



4.2.2.4 Alternative Medicine Advisor



4.3 Database Schema

- ***User Entity:***

  User(Uid, Name, Password, IsPharmacist, Email, Password)

- ***Inventory Entity(1 isolated table per user):***

  Inventory_Uid(Iname, Bid, Quantity, Purchase_Price, Sale_Price, MRP, Exp_Date, Purchase_Date, Location, Category)

  Mapping table for multivalued *Composition* attribute(exists only if user is Pharmacist):

  Composition(Iname, Component)

- ***Sales Entity(1 isolated table per user):***

  Sales_Uid(Bid (FK references Inventory_Uid), Month, Year, Sold, Quantity, Expenditure, Income)

4.4 Security Design

4.4.1 Authentication and Authorization Mechanisms

The system ensures secure access through a robust authentication and authorization framework. User passwords are encrypted using the Bcrypt hashing algorithm before being stored in the MySQL database.

Upon successful login, the Python Flask backend generates a JSON Web Token (JWT). This token is managed by the JavaScript frontend and used to statelessly authenticate subsequent requests, ensuring users can only access endpoints and data explicitly authorized for their specific role.

4.4.1 Security Protocols

The Smart Inventory and Sales Analytics System employs comprehensive security protocols to safeguard user data. To protect data in transit, all communication between the web browser and the Flask backend is strictly encrypted using HTTPS/TLS protocols.For data at rest, the system utilizes a cloud-hosted MySQL database, where infrastructural security is managed by the external hosting provider.

At the application level, data integrity is maintained through robust server-side access controls. To defend against common web vulnerabilities, the system implements rigorous input validation and parameterized queries to prevent SQL injection attacks. Furthermore, standard security measures and data sanitization are enforced to mitigate Cross-Site Scripting (XSS) and Cross-Site Request Forgery (CSRF) threats.

5.0. Traceability

4.1 Requirement-to-Design Mapping

Requirement/ Use Case ID	Requirement/ Use Case Name	Design ID(s)	Design Artifact
3.2.1	Inventory Management	D-CL-02, D-CL-04, D-SQ-01	Class Diagram(Inventory, Composition), Sequence Diagram(Inventory Management)
3.2.2	Sales Management	D-CL-02, D-CL-03, D-SQ-02	Class Diagrams(Inventory, Sales) Sequence Diagram(Sales Management)
3.2.3	Sales Analysis	D-CL-03, D-SQ-03	Class Diagram(Sales), Sequence Diagram(Sales Analysis)
3.2.4	Alternative Medicine Advisor	D-CL-02, D-CL-04, D-SQ-04	Class Diagram(Inventory, Composition), Sequence Diagram(Alternative Medicine Advisor)
3.3.1	Logical Structure of the Data	D-DB-01	Multi-Tenant Database Schema
3.3.2	Security	D-SEC-01, D-SEC-02	JWT Authentication, Bcrypt Password Encryption, Server-side Security Protocols

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Pharmacist, 2, 4, 5, 8, 9, 10, 11, 15, 16
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